

Project Title
Smart Recipe Discovery System Using Ingredients & Keywords

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
A. Amulya - 2520030032
P. Bhavana Manaswini - 2520030154
Ch. Gayatri Sri Harshita - 2520030600

Team Number: 13
Section: 8

Under the guidance of

Dr. V. Sireesha
Professor, Department of CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

In today's fast-paced lifestyle, deciding what to cook with the ingredients available at home has become a common challenge for many individuals and families. People often have different ingredients in their kitchens but may not know how to combine them to prepare a suitable meal. Traditional recipe platforms usually require users to search for a specific dish name, which can be inconvenient and time-consuming when users are unsure about what to cook. This may result in unused ingredients, food wastage, and unnecessary expenses.

The **Smart Recipe Discovery System** is developed to provide a simple, intelligent, and user-friendly solution to this problem. The system allows users to enter the ingredients they already have at home or search using simple keywords. Based on the provided input, the system identifies and recommends suitable recipes that match the available ingredients. This helps users discover different meal options without manually searching through a large number of recipes.

The system also provides filtering options to make recipe discovery more convenient. Users can refine their results based on cuisine type, preparation time, dietary preferences, meal category, and other requirements. Each recommended recipe can provide useful information such as required ingredients, cooking time, preparation steps, and serving details. This allows users to quickly select a recipe according to their preferences and available resources.

An important objective of the system is to promote the effective utilization of available food ingredients. By suggesting recipes based on existing ingredients, the system helps users use items that might otherwise remain unused or expire. This can reduce household food wastage while saving time, effort, and money. It also supports better meal planning by providing multiple recipe choices based on the user's needs.

Overall, the **Smart Recipe Discovery System** provides a practical and efficient approach to recipe discovery and everyday meal planning. It makes cooking decisions easier, encourages better ingredient utilization, and provides personalized recipe recommendations. The system can therefore serve as a useful digital solution for individuals and families seeking convenient, flexible, and efficient cooking options.

Signature of the Guide